A time–temperature scaling relation for isothermal sintering by vapor transport

enceladus_DSM `studies/keff_sintering/temperature_scaling/`

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1 Statement of the result

Consider an aggregate of ice grains of characteristic radius $\bar{R}$ in air, held at a uniform temperature T with no imposed gradient, evolving by sublimation, vapor diffusion and desublimation as described by the phase-field model of the manuscript. Let Q be any quantity determined by the geometry of the ice (specific surface area, a neck width, the effective conductivity divided by that of ice). Then

$$\boxed{Q(t;\,T) = \widehat{Q}\!\left(\frac{t}{\tau(T)};\;\Lambda(T)\right), \qquad \tau(T) = \frac{\bar{R}^2\,\beta(T)}{d_0(T)}, \qquad \Lambda(T) = \frac{L^*(T)}{\bar{R}}} \tag{1.1}$$

where $\widehat{Q}$ depends on the initial geometry but not otherwise on temperature. The time scale τ contains the saturation vapor density and changes by a factor of 29 between -40 and $-5\,^\circ$C; the group Λ contains the vapor diffusivity and changes by 20%. To the extent that this weak dependence can be ignored,

$$Q(t;\,T) = Q\!\left(\frac{t}{a_T};\;T_{\mathrm{ref}}\right), \qquad a_T = \frac{\tau(T)}{\tau(T_{\mathrm{ref}})} \simeq \sqrt{\frac{T}{T_{\mathrm{ref}}}}\;\frac{\rho^I_{vs}(T_{\mathrm{ref}})}{\rho^I_{vs}(T)}, \tag{1.2}$$

that is, a change of temperature is a change of the unit of time. Sections 2–7 derive this; Section 8 states it with its numbers; Section 9 gives the conditions under which it holds.

Temperatures in formulas are absolute. Symbols follow the manuscript: ϕ for the phase field, χ for curvature, subscripts "ice" and "air".

2 The model

The unknowns are the phase fields ϕ_{ice} and $\phi_{\mathrm{air}} = 1-\phi_{\mathrm{ice}}$, the temperature T and the vapor density ρ_v. The governing equations (manuscript Eq. 1) are

$$\frac{\partial \phi_{\mathrm{ice}}}{\partial t} = -3M_{\mathrm{sub}}\left[\frac{1}{\varepsilon}\,\phi_{\mathrm{ice}}(1-\phi_{\mathrm{ice}})(1-2\phi_{\mathrm{ice}}) - \varepsilon\nabla^2\phi_{\mathrm{ice}}\right] + \alpha_{\mathrm{sub}}\,\phi^2_{\mathrm{ice}}\phi^2_{\mathrm{air}}\,\frac{\rho_v - \rho^I_{vs}(T)}{\rho_{\mathrm{ice}}}, \tag{2.1}$$

$$\rho c_p\,\frac{\partial T}{\partial t} = \nabla\cdot(K\nabla T) - \rho\,L_{\mathrm{sub}}\,\frac{\partial \phi_{\mathrm{air}}}{\partial t}, \tag{2.2}$$

$$\frac{1}{\xi_v}\,\frac{\partial(\phi_{\mathrm{air}}\rho_v)}{\partial t} = \nabla\cdot(\phi_{\mathrm{air}}D_v\nabla\rho_v) + \rho_{\mathrm{ice}}\,\frac{\partial \phi_{\mathrm{air}}}{\partial t}. \tag{2.3}$$

Equation (2.3) is written as the solver implements it (`src/assembly.c`): the factor $\xi_v = 10^{-3}$ multiplies the diffusion and source terms together. With $\xi_v = 1$ it is manuscript Eq. (1c).

The kinetic parameters of Eq. (2.1) are set from physical ones (`src/material_properties.c`, following Moure & Fu, 2024):

$$\lambda_{\mathrm{sub}} = \frac{a_1\,\varepsilon\,\rho_{\mathrm{ice}}}{d_0\,\rho_{vs}^I}, \qquad \tau_{\mathrm{sub}} = \frac{\varepsilon^2\,(\beta_{\mathrm{sub}} + \Delta\beta)}{d_0}, \qquad M_{\mathrm{sub}} = \frac{\varepsilon}{3\,\tau_{\mathrm{sub}}}, \qquad \alpha_{\mathrm{sub}} = \frac{\lambda_{\mathrm{sub}}}{\tau_{\mathrm{sub}}}, \tag{2.4}$$

with $a_1 = 5$. The physical inputs are

$$d_0 = \frac{\gamma\, m}{\rho_{\mathrm{ice}}\, k_B T} \qquad \text{capillary length,} \tag{2.5}$$

$$\beta_{\mathrm{sub}} = \frac{1}{\alpha_c}\,\frac{\rho_{\mathrm{ice}}}{\rho_{vs}^I(T)}\sqrt{\frac{2\pi m}{k_B T}} \qquad \text{kinetic coefficient,} \tag{2.6}$$

$$\Delta\beta = a_1 a_2\,\varepsilon\left(\frac{1}{D_T} + \frac{1}{D_v(T)}\right)\frac{\rho_{\mathrm{ice}}}{\rho_{vs}^I(T)} \qquad \text{thin-interface term, } a_2 = 0.1581, \tag{2.7}$$

$$D_v = D_{v0}\left(\frac{T}{T_0}\right)^{1.81} \qquad \text{vapor diffusivity,} \tag{2.8}$$

and $\rho_{vs}^I(T)$, the saturation vapor density over flat ice (manuscript Eqs. 2–3). The condensation coefficient α_c is a constant.

Temperature enters through $\rho_{vs}^I(T)$, $D_v(T)$, $d_0 \propto 1/T$ and the explicit $\sqrt{T}$ in β_{sub}. Densities, heat capacities and conductivities are constants in the model.

3 Assumptions

A1. **Isothermal setting.** The mean temperature is uniform and constant, and there is no imposed gradient. Write $T = T_m + \vartheta$ with ϑ the departure caused by latent heat. (Its size is found in Section 6; it is not assumed to vanish.)

A2. **Sharp-interface limit.** The interface parameter is small compared with the radii of curvature, $\varepsilon \ll \bar{R}$, so the phase-field equation (2.1) may be replaced by its sharp-interface limit.

A3. **Constant α_c.** The condensation coefficient does not depend on temperature or supersaturation.

Three further simplifications are *not* assumed. They are obtained in Section 6 by showing that the corresponding dimensionless groups are small.

4 The sharp-interface problem

Let $\Gamma(t)$ be the ice surface, $\mathbf{n}$ its unit normal pointing from the ice into the pore, and v_n its velocity along $\mathbf{n}$, so that $v_n > 0$ when ice grows.

4.1 Vapor in the pore

In the pore $\phi_{\text{air}} = 1$ and $\partial\phi_{\text{air}}/\partial t = 0$, and at uniform temperature D_v is uniform. Equation (2.3) becomes

$$\frac{1}{\xi_v}\,\frac{\partial \rho_v}{\partial t} = D_v\,\nabla^2 \rho_v \qquad \text{in the pore.} \tag{4.1}$$

In the ice $\phi_{\text{air}} = 0$ and Eq. (2.3) is empty.

4.2 Mass balance at the surface

Take a coordinate n along $\mathbf{n}$ with the surface at $n = s(t)$, $\mathrm{d}s/\mathrm{d}t = v_n$. Across the surface ϕ_{air} steps from 0 to 1, $\phi_{\text{air}} = H(n - s(t))$ with H the unit step, so

$$\frac{\partial \phi_{\text{air}}}{\partial t} = -v_n\,\delta(n-s), \qquad \frac{\partial(\phi_{\text{air}}\rho_v)}{\partial t} = -v_n\,\rho_v\,\delta(n-s) + \phi_{\text{air}}\,\frac{\partial \rho_v}{\partial t}.$$

Integrate Eq. (2.3) over n from $s-\eta$ to $s+\eta$ and let $\eta \to 0$. The bounded term $\phi_{\text{air}}\,\partial\rho_v/\partial t$ contributes nothing. The flux term gives $\phi_{\text{air}} D_v\,\partial\rho_v/\partial n$ evaluated on the pore side (it vanishes on the ice side, where $\phi_{\text{air}} = 0$). Hence

$$-\frac{1}{\xi_v}\,v_n\,\rho_v = D_v\,\left.\frac{\partial \rho_v}{\partial n}\right|_{\Gamma} - \rho_{\text{ice}}\,v_n,$$

that is,

$$D_v\,\left.\frac{\partial \rho_v}{\partial n}\right|_{\Gamma} = \left(\rho_{\text{ice}} - \frac{\rho_v}{\xi_v}\right) v_n. \tag{4.2}$$

This is conservation of water at the surface: ice that grows ($v_n > 0$) is supplied by vapor diffusing toward it, so ρ_v increases with distance from a growing surface.

4.3 Kinetics at the surface

The matched-asymptotic analysis of Eq. (2.1) for $\varepsilon \ll \bar{R}$ (Kaempfer & Plapp, 2009; Moure & Fu, 2024), with the parameters of Eq. (2.4), gives the Gibbs–Thomson condition

$$\frac{\rho_v - \rho_{vs}^I(T_\Gamma)}{\rho_{vs}^I(T_\Gamma)} = d_0\,\chi + \beta\,v_n \quad \text{on } \Gamma, \qquad \beta \equiv \beta_{\text{sub}} + \Delta\beta = \beta_{\text{sub}}\,(1+\delta), \tag{4.3}$$

where χ is the curvature of the surface (positive for a convex grain), T_Γ the local surface temperature, and

$$\delta \equiv \frac{\Delta\beta}{\beta_{\text{sub}}} = a_1 a_2\,\varepsilon\,\alpha_c \left(\frac{1}{D_T} + \frac{1}{D_v}\right)\sqrt{\frac{k_B T}{2\pi m}}. \tag{4.4}$$

This step is quoted, not derived here: it is the property the model is constructed to have, and the cited papers give the analysis. The factor $\rho_{\text{ice}}/\rho_{vs}^I$ is common to β_{sub} and $\Delta\beta$ and cancels in δ.

4.4 Heat

Away from the surface Eq. (2.2) is conduction in each phase,

$$\rho_j c_{p,j} \frac{\partial \vartheta}{\partial t} = k_j \nabla^2 \vartheta, \qquad j \in \{\text{ice},\ \text{air}\}. \tag{4.5}$$

Integrating Eq. (2.2) across the surface in the same way as in Section 4.2, with $\partial \phi_{\text{air}}/\partial t = -v_n\, \delta(n-s)$ and the density of the phase that forms, ρ_{ice} (Eq. 2.2 weights this term with the mixture density; the difference is immaterial here because the group it feeds, G_T below, is negligible),

$$k_{\text{ice}} \left.\frac{\partial \vartheta}{\partial n}\right|_{\text{ice}} - k_{\text{air}} \left.\frac{\partial \vartheta}{\partial n}\right|_{\text{air}} = \rho_{\text{ice}}\, L_{\text{sub}}\, v_n, \tag{4.6}$$

with ϑ continuous. Growth releases latent heat, which is conducted away into both phases.

4.5 Coupling of heat to the kinetics

The surface temperature enters Eq. (4.3) through $\rho_{vs}^I(T_\Gamma)$. With $T_\Gamma = T_m + \vartheta$ and $|\vartheta| \ll T_m$,

$$\rho_{vs}^I(T_m + \vartheta) = \rho_{vs}^I(T_m)\left[1 + s\,\vartheta + O(\vartheta^2)\right], \qquad s \equiv \left.\frac{\mathrm{d} \ln \rho_{vs}^I}{\mathrm{d}T}\right|_{T_m} \simeq \frac{L_{\text{sub}}\, m}{k_B T_m^2},$$

the last form being the Clausius–Clapeyron relation ($s = 0.096\ \mathrm{K}^{-1}$ at $-20\,^\circ\mathrm{C}$). Define the supersaturation relative to flat ice at the mean temperature,

$$\sigma \equiv \frac{\rho_v - \rho_{vs}^I(T_m)}{\rho_{vs}^I(T_m)}. \tag{4.7}$$

Then, to first order in the small quantities σ and $s\vartheta$,

$$\frac{\rho_v - \rho_{vs}^I(T_\Gamma)}{\rho_{vs}^I(T_\Gamma)} = \frac{1+\sigma}{1+s\vartheta} - 1 = \sigma - s\,\vartheta,$$

and Eq. (4.3) becomes

$$\sigma = d_0\, \chi + \beta\, v_n + s\,\vartheta \qquad \text{on } \Gamma, \tag{4.8}$$

with d_0 and β evaluated at T_m. From here on ρ_{vs}^I, d_0, β, D_v denote values at T_m, and T is written for T_m.

4.6 Summary of the dimensional problem

Unknowns: ρ_v (equivalently σ) in the pore, ϑ everywhere, and the surface $\Gamma(t)$. Equations: (4.1), (4.2), (4.8), (4.5), (4.6), with periodic boundaries.

5 Scales and dimensionless variables

Choose

$$\begin{aligned} &\mathbf{x} = \bar{R}\,\mathbf{x}', && \chi = \frac{\chi'}{\bar{R}}, && \sigma = \frac{d_0}{\bar{R}}\,u, \\ &v_n = V\, v', \quad V \equiv \frac{d_0}{\bar{R}\,\beta}, && t = \tau\, t', \quad \tau \equiv \frac{\bar{R}}{V} = \frac{\bar{R}^2\beta}{d_0}, && \vartheta = \Theta\,\vartheta', \quad \Theta \equiv \frac{\rho_{\text{ice}} L_{\text{sub}} V \bar{R}}{k_{\text{ice}}}. \end{aligned} \tag{5.1}$$

The reasons:

- $\bar{R}$ is the only length in the geometry.
- $d_0/\bar{R}$ is the supersaturation at which a grain of radius $\bar{R}$ is in equilibrium (Eq. 4.8 with $v_n = 0$, $\vartheta = 0$, $\chi = 1/\bar{R}$), so u is of order one.
- V is the velocity that the kinetic term βv_n needs for it to be as large as the capillary term $d_0/\bar{R}$.
- τ is the time for a surface moving at V to travel $\bar{R}$.
- Θ is the temperature difference that conducts the latent heat released at velocity V across a distance $\bar{R}$ of ice.

Derivatives transform as

$$\nabla = \frac{1}{\bar{R}}\nabla', \qquad \frac{\partial}{\partial n} = \frac{1}{\bar{R}}\frac{\partial}{\partial n'}, \qquad \frac{\partial}{\partial t} = \frac{1}{\tau}\frac{\partial}{\partial t'},$$

and from Eq. (4.7), $\rho_v = \rho_{vs}^I \left(1 + \dfrac{d_0}{\bar{R}} u\right)$, so that $\nabla\rho_v = \rho_{vs}^I \dfrac{d_0}{\bar{R}^2}\nabla' u$ and $\partial\rho_v/\partial t = \rho_{vs}^I \dfrac{d_0}{\bar{R}\,\tau}\,\partial u/\partial t'$.

6 Nondimensionalization

6.1 Kinetics

Substituting (5.1) in Eq. (4.8):

$$\frac{d_0}{\bar{R}}\,u = d_0\,\frac{\chi'}{\bar{R}} + \beta\,\frac{d_0}{\bar{R}\,\beta}\,v' + s\,\Theta\,\vartheta'.$$

Dividing by $d_0/\bar{R}$,

$$u = \chi' + v' + G_T\,\vartheta', \qquad G_T \equiv \frac{s\,\Theta\,\bar{R}}{d_0} = \frac{s\,\rho_{\rm ice} L_{\rm sub}\,\bar{R}}{k_{\rm ice}\,\beta}, \tag{6.1}$$

where $\Theta\bar{R}/d_0 = \rho_{\rm ice}L_{\rm sub}V\bar{R}^2/(k_{\rm ice}d_0) = \rho_{\rm ice}L_{\rm sub}\bar{R}/(k_{\rm ice}\beta)$ by the definition of V.

6.2 Mass balance

The left side of Eq. (4.2) is $D_v\,\rho_{vs}^I\,(d_0/\bar{R}^2)\,\partial u/\partial n'$. On the right, $\rho_v = \rho_{vs}^I(1 + O(d_0/\bar{R}))$, so

$$D_v\,\rho_{vs}^I\,\frac{d_0}{\bar{R}^2}\,\frac{\partial u}{\partial n'} = \rho_{\rm ice}\left(1 - \frac{\rho_{vs}^I}{\xi_v\,\rho_{\rm ice}}\right) V\,v'.$$

Solving for v' and inserting $V = d_0/(\bar{R}\beta)$,

$$v' = \frac{D_v\,\rho_{vs}^I\,d_0}{\rho_{\rm ice}\,\bar{R}^2}\cdot\frac{\bar{R}\,\beta}{d_0}\cdot\frac{1}{1-m_v}\,\frac{\partial u}{\partial n'} = \frac{D_v\,\beta\,\rho_{vs}^I}{\rho_{\rm ice}\,\bar{R}}\cdot\frac{1}{1-m_v}\,\frac{\partial u}{\partial n'}, \qquad m_v \equiv \frac{\rho_{vs}^I}{\xi_v\,\rho_{\rm ice}}.$$

Define the length

$$L^* \equiv \frac{D_v\,\beta\,\rho_{vs}^I}{\rho_{\rm ice}} = (1+\delta)\,\frac{D_v(T)}{\alpha_c}\sqrt{\frac{2\pi m}{k_B T}}, \tag{6.2}$$

the second form following from Eqs. (2.6) and (4.3): $\beta\,\rho_{vs}^I/\rho_{\rm ice} = (1+\delta)\,\alpha_c^{-1}\sqrt{2\pi m/(k_BT)}$. **The saturation vapor density cancels in L^*.** With $\Lambda \equiv L^*/\bar{R}$,

$$v' = \frac{\Lambda}{1-m_v}\,\frac{\partial u}{\partial n'} \qquad \text{on } \Gamma. \tag{6.3}$$

6.3 Vapor diffusion

Substituting in Eq. (4.1), both sides carry $\rho_{vs}^I\, d_0/\bar{R}$:

$$\frac{1}{\xi_v}\,\rho_{vs}^I\,\frac{d_0}{\bar{R}\,\tau}\,\frac{\partial u}{\partial t'} = D_v\,\rho_{vs}^I\,\frac{d_0}{\bar{R}^3}\,\nabla'^2 u \quad\Longrightarrow\quad \frac{P}{\xi_v}\,\frac{\partial u}{\partial t'} = \nabla'^2 u, \qquad P \equiv \frac{\bar{R}^2}{D_v\,\tau} = \frac{d_0}{D_v\,\beta}.$$

P is the time for vapor to diffuse a distance $\bar{R}$ divided by the time for the geometry to change.

6.4 Heat

Equation (4.5) becomes $(\bar{R}^2/(D_{T,j}\,\tau))\,\partial\vartheta'/\partial t' = \nabla'^2\vartheta'$ with $D_{T,j} = k_j/(\rho_j c_{p,j})$, and Eq. (4.6), divided by $k_{\text{ice}}\Theta/\bar{R} = \rho_{\text{ice}} L_{\text{sub}} V$,

$$\left.\frac{\partial\vartheta'}{\partial n'}\right|_{\text{ice}} - \frac{k_{\text{air}}}{k_{\text{ice}}}\left.\frac{\partial\vartheta'}{\partial n'}\right|_{\text{air}} = v'.$$

So ϑ' is of order one, and it enters the kinetics only through the term $G_T\,\vartheta'$ of Eq. (6.1).

6.5 Sizes of the groups

Evaluated for the aggregate runs ($\bar{R} = 50$ μm, $\varepsilon = 1$ μm, $\alpha_c = 1.020 \times 10^{-3}$) by `check_groups.py`:

Group	Meaning	−40 °C	−30 °C	−20 °C	−10 °C	−5 °C
$\Lambda = L^*/\bar{R}$	attachment against diffusion	2.50	2.64	2.78	2.93	3.00
δ	thin-interface term	0.0200	0.0200	0.0199	0.0199	0.0199
G_T	latent heat on saturation	1×10^{-4}	3×10^{-4}	7×10^{-4}	2×10^{-3}	2×10^{-3}
$m_v = \rho_{vs}^I/(\xi_v \rho_{\text{ice}})$	vapor mass in the balance	1×10^{-4}	3×10^{-4}	9×10^{-4}	2×10^{-3}	4×10^{-3}
P/ξ_v	vapor storage	1×10^{-9}	3×10^{-9}	7×10^{-9}	2×10^{-8}	2×10^{-8}
$\bar{R}^2/(D_T\tau)$	heat storage	10^{-12} to 10^{-10}				

G_T, m_v, P/ξ_v and the heat-storage group are all below 4×10^{-3}. Setting them to zero:

- $P/\xi_v \to 0$: the vapor field is quasi-steady, $\nabla'^2 u = 0$.
- $m_v \to 0$: the mass balance is $v' = \Lambda\,\partial u/\partial n'$.
- $G_T \to 0$: the heat problem decouples from the kinetics, and the evolution is that of an isothermal system. (The scale Θ is 2×10^{-8} to 6×10^{-7} K.)

δ is not small compared with these, but it is the same at every temperature to three digits, so it rescales β by a constant and has already been absorbed in τ and L^*.

7 The reduced problem

$$\boxed{\begin{aligned} \nabla'^2 u &= 0 && \text{in the pore,} \\ u &= \chi' + v' && \text{on } \Gamma', \\ v' &= \Lambda\,\frac{\partial u}{\partial n'} && \text{on } \Gamma', \end{aligned}} \tag{7.1}$$

with periodic boundaries, the surface Γ' moving at normal velocity v' in the time $t' = t/\tau$. Eliminating v', the two surface conditions are the single condition $u - \Lambda\,\partial u/\partial n' = \chi'$.

Problem (7.1) contains one parameter, Λ. Given the initial geometry in units of $\bar{R}$, its solution $\Gamma'(t')$ is a function of t' and Λ only.

Meaning of Λ. A vapor molecule travelling a distance ℓ between two surfaces meets a diffusive resistance proportional to ℓ and an attachment resistance proportional to L^*. For $\Lambda \to \infty$ the field u is uniform, $v' = \bar{u} - \chi'$ with $\bar{u}$ fixed by conservation of ice ($\oint v'\,\mathrm{d}\Gamma' = 0$), and the problem has no parameter at all. For $\Lambda \to 0$, $u = \chi'$ on the surface and diffusion alone limits the rate. The aggregates have $\Lambda \approx 2.5$ to 3: attachment is the larger resistance for transport between neighboring surfaces, and diffusion is not negligible.

8 The time–temperature scaling relation

8.1 Statement

Because Problem (7.1) determines Γ' as a function of t/τ and Λ, any geometric quantity Q, made dimensionless with $\bar{R}$, satisfies Eq. (1.1):

$$Q(t;\,T) = \widehat{Q}\left(\frac{t}{\tau(T)};\ \Lambda(T)\right).$$

This applies to the specific surface area (as $\mathrm{SSA}\cdot\bar{R}$), to neck widths, and to $k_{\mathrm{eff}}/k_{\mathrm{ice}}$, which for fixed $k_{\mathrm{air}}/k_{\mathrm{ice}}$ is a functional of the geometry alone.

8.2 The time scale

From Eqs. (5.1), (2.5), (2.6) and (4.3),

$$\tau = \frac{\bar{R}^2\beta}{d_0} = \bar{R}^2\cdot\frac{1+\delta}{\alpha_c}\,\frac{\rho_{\mathrm{ice}}}{\rho_{vs}^I}\sqrt{\frac{2\pi m}{k_BT}}\cdot\frac{\rho_{\mathrm{ice}}\,k_BT}{\gamma\,m},$$

and since $\dfrac{k_BT}{m}\sqrt{\dfrac{2\pi m}{k_BT}} = \sqrt{\dfrac{2\pi k_BT}{m}}$,

$$\boxed{\tau(T) = (1+\delta)\,\frac{\bar{R}^2\,\rho_{\mathrm{ice}}^2}{\alpha_c\,\gamma}\,\frac{1}{\rho_{vs}^I(T)}\sqrt{\frac{2\pi k_BT}{m}}} \tag{8.1}$$

The temperature dependence is $\tau \propto \sqrt{T}/\rho_{vs}^I(T)$.

8.3 The shift factor

Relative to a reference temperature,

$$a_T \equiv \frac{\tau(T)}{\tau(T_{\mathrm{ref}})} = \sqrt{\frac{T}{T_{\mathrm{ref}}}}\,\frac{\rho_{vs}^I(T_{\mathrm{ref}})}{\rho_{vs}^I(T)}, \tag{8.2}$$

the factor $(1+\delta)$ cancelling to 10^{-4}. It does not depend on $\bar{R}$, α_c, γ or ε.

	$-40\,^\circ$C	$-30\,^\circ$C	$-20\,^\circ$C	$-10\,^\circ$C	$-5\,^\circ$C
ρ_{vs}^I [10^{-4} kg m^{-3}]	1.055	3.123	8.487	21.39	33.11
V [μm d^{-1}]	0.029	0.083	0.221	0.546	0.838
τ [d]	1747	603	226	91.5	59.7
a_T ($T_{\mathrm{ref}} = -20\,^\circ$C)	7.72	2.66	1	0.404	0.264
L^* [μm]	125.0	132.0	139.2	146.4	150.1
Λ	2.50	2.64	2.78	2.93	3.00

Between -40 and $-5\,^\circ$C, τ changes by a factor of 29.3: a factor of 31.4 from ρ_{vs}^I and 0.93 from $\sqrt{T}$.

8.4 Activation energy

In the model ρ_{vs}^I is proportional to the saturation vapor pressure (for $P_{vs} \ll P_{\mathrm{atm}}$), which obeys the Clausius–Clapeyron relation, so $\rho_{vs}^I \propto \exp(-L_{\mathrm{sub}} M_w/(R_g T))$ with M_w the molar mass and R_g the gas constant. Then, from Eq. (8.1),

$$E_a \equiv R_g \frac{\mathrm{d}\ln\tau}{\mathrm{d}(1/T)} = L_{\mathrm{sub}}\, M_w - \tfrac{1}{2} R_g T \approx 50\ \mathrm{kJ\,mol^{-1}}. \tag{8.3}$$

The formula gives 49.9 at $-20\,^\circ$C; a fit of $\ln\tau$ against $1/T$ over the five temperatures gives 50.1.

8.5 The residual dependence

From Eq. (6.2) and (2.8), $L^* \propto D_v/\sqrt{T} \propto T^{1.31}$, so Λ changes by $(268.15/233.15)^{1.31} = 1.20$ across the range. This is the only way the vapor diffusivity enters. If the solution of Problem (7.1) is insensitive to a 20% change in Λ about $\Lambda \approx 2.8$, Eq. (1.1) reduces to Eq. (1.2): curves at different temperatures coincide when plotted against t/a_T. In the limit $\Lambda \to \infty$ the coincidence is exact.

8.6 A consequence that does not involve time

Eliminating t/τ between two geometric quantities Q_1 and Q_2 of the same run gives a relation $Q_1 = F(Q_2; \Lambda)$ that does not contain τ at all. Temperature enters it only through Λ. This is why k_{eff} plotted against SSA for one packing is the same at every temperature, and why that comparison can be tighter than the comparison of rates.

8.7 Relation to the clock used in the manuscript

From Eq. (2.4),

$$\tau_{\mathrm{sub}} = \frac{\varepsilon^2 \beta}{d_0} = \left(\frac{\varepsilon}{\bar{R}}\right)^2 \tau, \qquad \theta \equiv \frac{t}{\tau_{\mathrm{sub}}} = \left(\frac{\bar{R}}{\varepsilon}\right)^2 \frac{t}{\tau}. \tag{8.4}$$

With $\varepsilon/\bar{R} = 1/50$ in every aggregate run, $\theta = 2500\, t/\tau$, so collapse against θ and collapse against t/τ are the same statement, and $\tau_{\mathrm{sub}}(T)/\tau_{\mathrm{sub}}(T_{\mathrm{ref}}) = a_T$. Check: 226.3 d/2500 = 7822 s, the value of τ_{sub} at $-20\,^\circ$C in manuscript Table 2; the other four temperatures agree to 0.03%.

9 Conditions of validity

The relation rests on A1–A3 and on the smallness of the groups of Section 6.5. It fails, or needs amending, when:

- α_c **depends on temperature.** Then τ acquires that dependence and so does Λ (Eq. 6.2); the first changes a_T, the second breaks the collapse if α_c changes much.
- **A second transport mechanism acts**, such as surface diffusion. It adds a term to the surface velocity with its own time scale and temperature dependence; the ratio of the two time scales is a new group that varies with temperature.
- **A temperature gradient is imposed.** ρ_{vs}^I then varies in space at order $s\,\bar{R}\,|\nabla T|\,\bar{R}/d_0$ in units of u, which exceeds one for gradients above about $4\ \mathrm{K\,m^{-1}}$ at these sizes. The isothermal problem no longer applies.
- Λ **is of order one or less and varies more**: larger grains, a larger α_c, or a wider temperature range. The diffusivity then matters in full. For the Molaro pairs ($\alpha_c = 0.1$, $\bar{R} \approx 87\ \mu$m) $\Lambda \approx 0.02$.
- **The interface is not thin**, $\varepsilon/\bar{R}$ not small compared with the local radius of curvature. The departure from the sharp-interface problem is the same at every temperature if $\varepsilon/\bar{R}$ is fixed, so it changes $\widehat{Q}$ but not the collapse.

10 Comparison with the simulations

The 125 aggregate runs (five packings at each of five porosities, each at five temperatures) test the relation with the full diffuse-interface model:

- Rates follow τ_{sub} to within 4%: a packing reaches a given SSA at times in the ratio a_T.
- k_{eff} at matched SSA agrees across the five temperatures to within 0.27% for every packing (median 0.07%), the time-free consequence of Section 8.6.

A 20% change in Λ is of the right size to account for the 4%; that it is the cause has not been tested. The analysis shows that the collapse is a property of the model for any starting geometry, and the simulations show how closely the diffuse-interface implementation follows it.

A Constants used

$\rho_{\mathrm{ice}} = 919\ \mathrm{kg\,m^{-3}}$; $k_{\mathrm{ice}} = 2.29$, $k_{\mathrm{air}} = 0.02\ \mathrm{W\,m^{-1}\,K^{-1}}$; $L_{\mathrm{sub}} = 2.83 \times 10^6\ \mathrm{J\,kg^{-1}}$; $\gamma = 0.109\ \mathrm{J\,m^{-2}}$; $m = 2.9915 \times 10^{-26}$ kg; $D_{v0} = 2.178 \times 10^{-5}\ \mathrm{m^2\,s^{-1}}$ at $T_0 = 273.15$ K; $D_T = 7.78 \times 10^{-6}\ \mathrm{m^2\,s^{-1}}$; $a_1 = 5$, $a_2 = 0.1581$; $\xi_v = 10^{-3}$. Every number in this note is produced by `check_groups.py`, which also asserts that the closed form (8.1) equals $\bar{R}^2\beta/d_0$ and that $(\varepsilon/\bar{R})^2\tau$ reproduces manuscript Table 2.